Production Information Systems

Chapter 7 :

Information Architecture and Logical Design

ISE Department

Year 4/ Level 7

Instructor : Dr. Jomana Bashatah

Outline

1. Introduction
2. IDEF Representation of ER Modeling
3. A Case Study on Data Modeling

2. IDEF Representation of ER Modeling

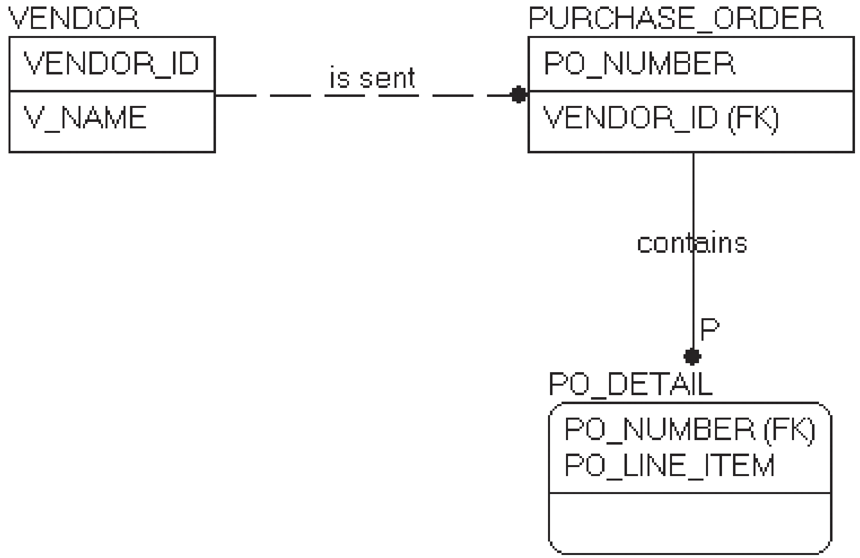
- E-R diagrams may contain elements (like composite attributes and non-1:M relationships) that need refinement before mapping to database tables
  - Data models should be in third normal form with composite attributes broken down before implementation
- IDEF1X is a variation of E-R diagrams that appears in commercial Computer Aided Software Engineering (CASE) tools
- IDEF1X complements IDEF0 (functional modeling methodology) by representing information required for IDEF0 functions
- IDEF1X models can be directly compiled into database tables through CASE tools
- "Informational architecture" refers to a refined data model that complements a related functional architecture

2. IDEF Representation of ER Modeling

- IDEF1X combines relational theory and entity-relationship modeling concepts
- IDEF1X is designed for effective communication between designers and users during conceptual design

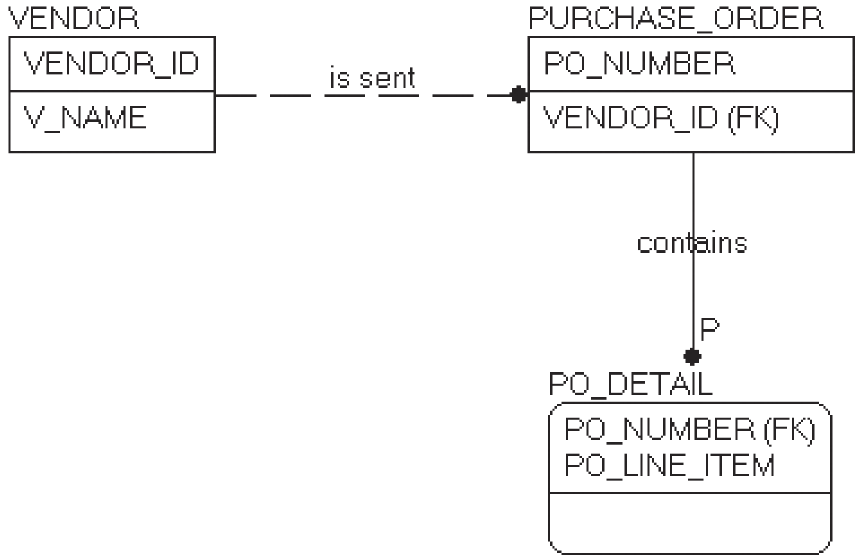
2. IDEF Representation of ER Modeling
2.1 IDEF1X Fundamentals

- In IDEF1X diagrams:
  - Entities are represented by boxes with the entity name (singular noun phrase) above
  - Attributes are contained within the entity box, making it more compact than generic E-R modeling
  - Primary key attributes appear in the upper portion, separated by a line from non-primary key attributes



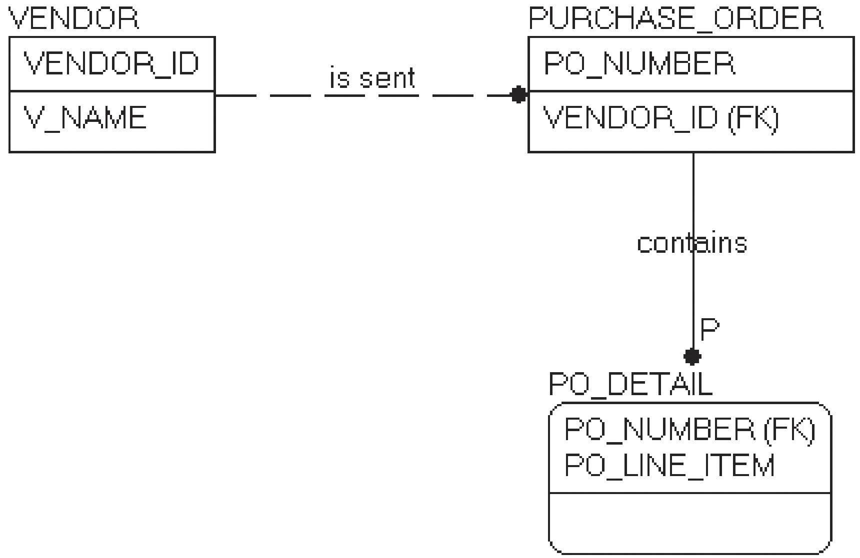
2. IDEF Representation of ER Modeling
2.1 IDEF1X Fundamentals

- IDEF1X distinguishes between Entity Types
  - Independent
  - Dependent
- Entity Types:
  - Independent entities (right-angle corners): Can be uniquely identified without other entities (e.g., VENDOR, PURCHASE_ORDER)
  - Dependent entities (rounded corners): Unique identification depends on relationship with another entity (e.g., PO_DETAIL)



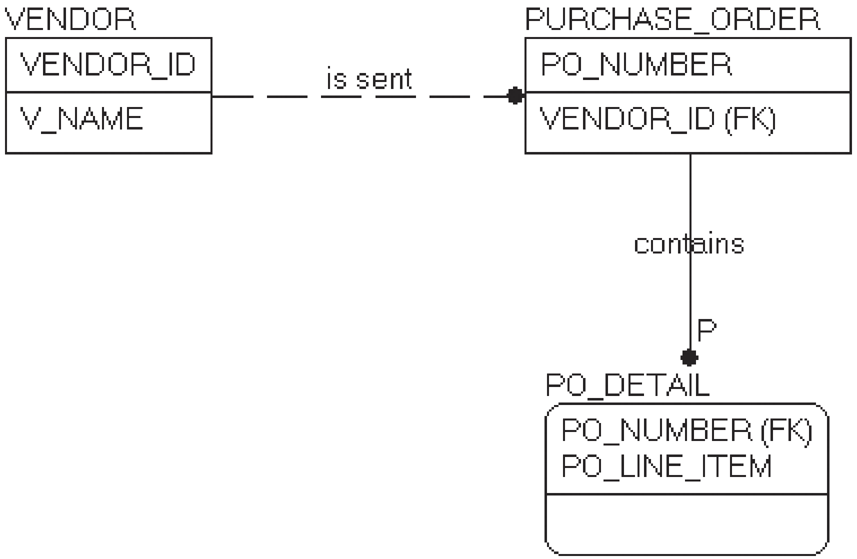
2. IDEF Representation of ER Modeling
2.1 IDEF1X Fundamentals

- Relationships:
  - Parent-child relationship: Associates one entity (parent) with zero, one, or more instances of another entity (child)
  - Each instance of the child entity is associated with exactly one instance of the parent entity
  - Shown as an arc from parent to child, with a dot at the child end
  - Named with a verb phrase placed beside the relationship line, expressed in parent-to-child direction
- Cardinality Notation:
  - Default is 1:M (one-to-many) unless specified
  - "Z" beside the dot indicates cardinality of zero or one
  - "P" beside the dot indicates cardinality of one or more
  - "n" beside the dot indicates cardinality of exactly n



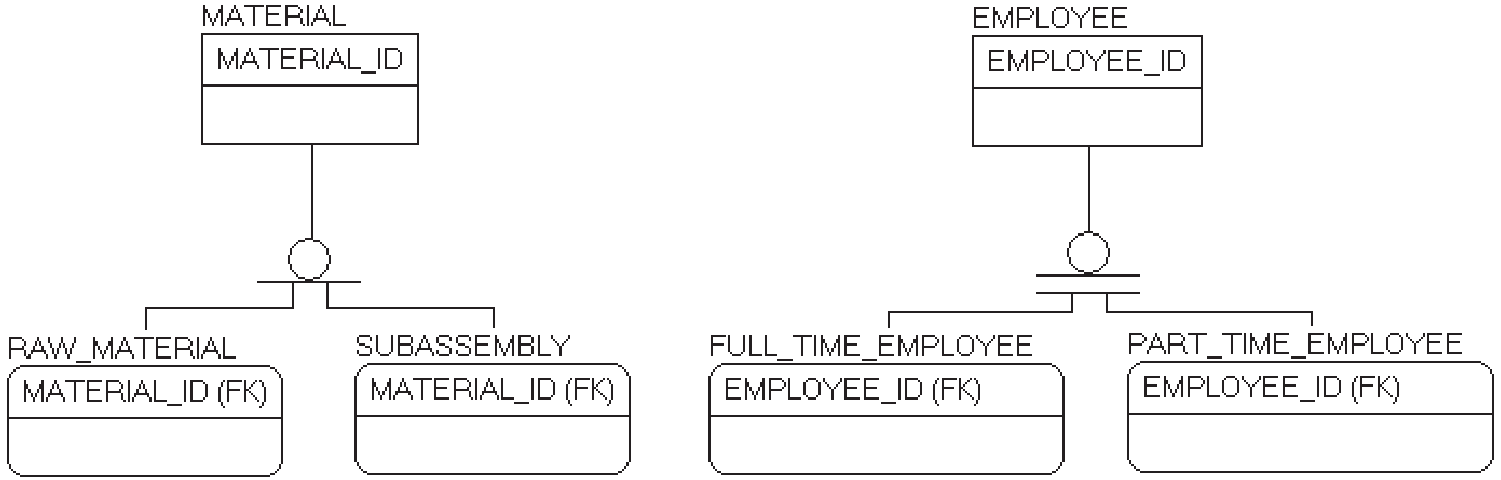
2. IDEF Representation of ER Modeling
2.1 IDEF1X Fundamentals

- Relationship Types:
  - Identifying relationship (solid line): Primary key attributes of parent become primary key attributes of child
  - Non-identifying relationship (dotted line): Foreign key is not part of the primary key of the M side
  - Optional relationship: Parent entity may not have relationship to child (shown by default dot with no notation)
  - Mandatory relationship: All instances must participate (shown with "P" next to dot)
  - Diamond on 1 side: Indicates M side may contain NULL values of foreign key (optional to child)
  - Note: optional relationship can only occur in a non-identifying relationship since all PKs in a parent become PKs in a child and PKs cannot be NULL

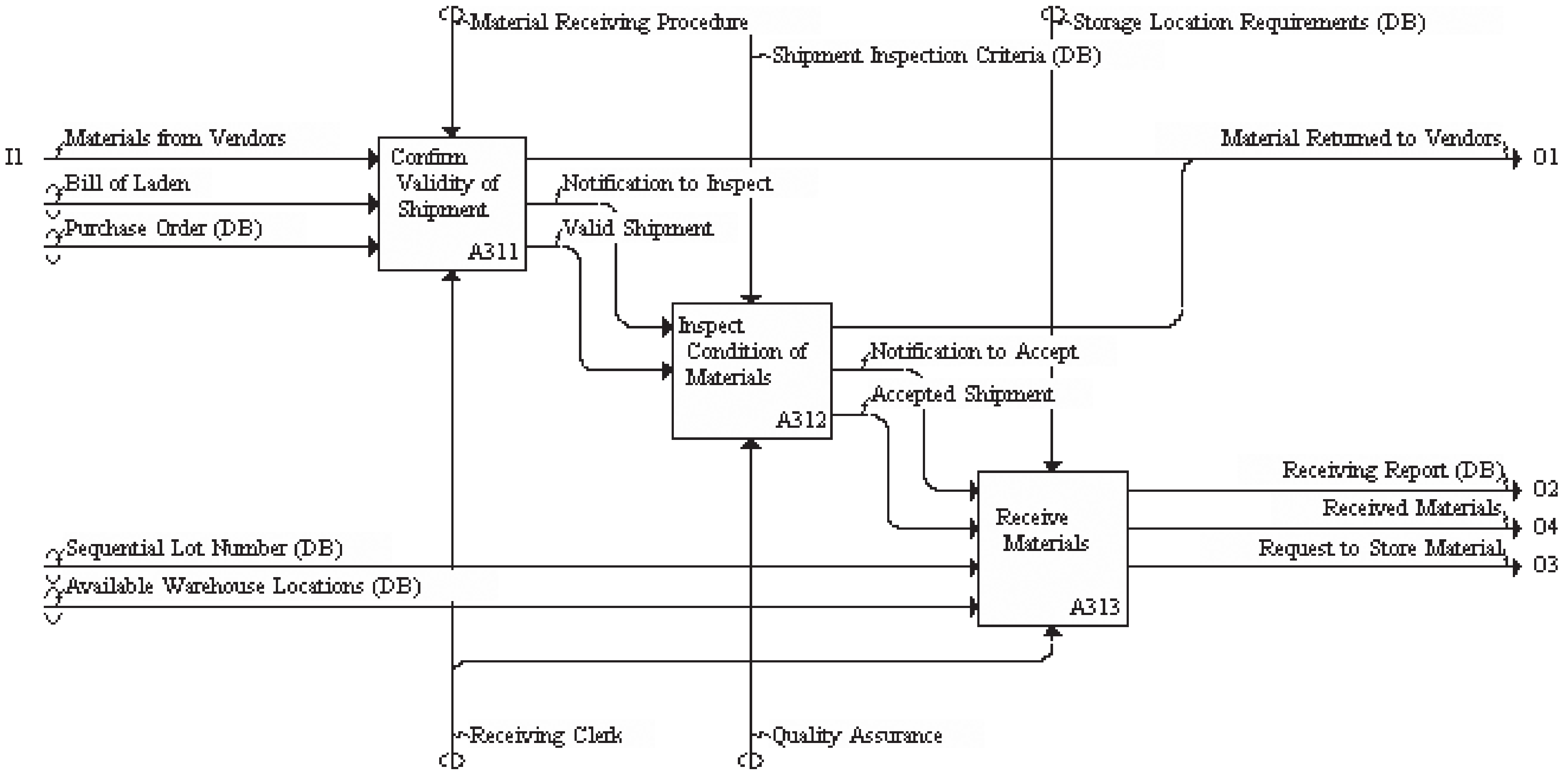


2. IDEF Representation of ER Modeling
2.1 IDEF1X Fundamentals

- Categorization Relationships (Superclass/Subclass):
  - Generic entity (superclass) connects to category entities (subclasses)
  - Category entities use the same primary key as the generic entity
  - An instance of generic entity relates to at most one instance of a category entity
  - Discriminator shows relationship nature (single/double bar with circle)
  - Incomplete categorization (single bar): Some generic instances don't appear in any category
  - Complete categorization (double bar with circle): All generic instances appear in categories
- IDEF1X supports 1:M, 1:1, and M:N relationships, but we will focus on 1:M relationships



3. A Case Study on Data Modeling



3. A Case Study on Data Modeling

- Refer to nodes A311 and A313. The receiving clerk is the initial contact when material arrives via truck at the receiving dock. The clerk is given a bill of laden by the truck driver, and he or she must confirm that the material being delivered is part of an open purchase order. Once it is determined that the material should be accepted, the truck is unloaded and the receiving clerk begins to account for the delivery by writing up a document called a receiving report. Therefore, three paper forms are being used at nodes A311 and A313:
  - the bill of laden
  - the purchase order
  - and the receiving report

3. A Case Study on Data Modeling

THE UNIVERSITY FOOD COMPANY
1766 NEW ENGLAND AVE.
PISCATAWAY, NJ 08855

VENDOR: General Provisions
125 Common St.
Boise, ID 44830
Attn: Dave Stauffer

P.O. Number: 3502
Date: 5/28/06
Amount: $4920.00

Line Item	Description	Unit Price	Amount	Del. Date
1	1400 gallons of tomato paste, Item # 2-414.	$3.40	$4760.00	6/25/06
	Freight (Est.)		160.00	
	Total		4920.00	

Ordered by: Randy Kim
Phone Number: 908-445-3657
Deliver to: The University Food Company
1766 New England Ave.
Piscataway, NJ 08855

FREIGHT BILL OF LADEN

To: The University Food Company
1766 New England Ave.
Piscataway, NJ 08855

From: General Provisions
125 Common St.
Boise, ID 44830

Date Shipped: Jun 20 06
Carrier: Northwest Trucking Co.
Hansford, WA
Car Truck Number: 5501
B/L No.: 06-225

UPC Code	Quantity	Weight	Description	Reference
20000 - 16605	1400	39,100	Gallon cans of Tomato paste	PO3502
	39	1,950	wood pallets	
	Total	41,050		

Received by: Jerry Dobbs
Date: June 25, 2006
Driver: Mack Truck
Lic. Number: E3589-002

RECEIVING REPORT

Supplier: General Provisions
125 Common St.
Boise, ID 44830

Purchase Order No.: PO3502
Date Received: June 25 2006

Quantity accepted	Quantity not accepted	Mfg. Lot No.	Item Code	Mat'L Lot No.	Description	Storage Location
1000		1275	RM805	97275	Tomato Paste, 1 gallon cans	Area A, Aisle 1 tier 1, bins 10-18
300		1283	"	97276	" " " "	Area A, Aisle 1 Tier 2, Bins 10-13
	100	"	"		" " " "	returned(1)

Comments: (1) returned due to case damage and badly dented containers.
Received by: J. Dobbs

3. A Case Study on Data Modeling

- The bill of laden is a shipping document that is used to identify the contents of the shipment and the shipping authority. The carrier presents the bill of laden when it arrives with a delivery.
- The hypothetical bill of laden has header information regarding the sending and receiving organizations as well as the shipper and date shipped. The body of the form identifies what is being shipped and a reference to the billing authority, in this case a purchase order number.
- The bottom of the form requires a signature by the individual who accepts the shipment, usually the shipping clerk. A name and driver's license number also identify the truck driver.
- The contents of a bill of laden will vary. However, we are going to assume that this is a generic form for our purposes in designing the data model for the function of receiving incoming materials.

3. A Case Study on Data Modeling

- For the purchase order, the MATERIAL_ID is not shown on this purchase order. The MATERIAL_ID is an internal identifier assigned by the enterprise to keep track of its own inventory. This number means nothing to the vendors, who have their own identifier code for the products they sell. Thus, the purchasing agent uses the description field on the purchase order to specify the product using the vendor's own identification, in this case "Item# 2-414."
- A company contact name and phone number is given at the bottom of the purchase order so that the vendor has someone to contact in the event she or he has to talk to a company representative. The contact is usually the purchasing agent initiating the purchase order. Also, a delivery address is given. A company may have more than one location that receives shipments; therefore, a shipping address is standard information on a purchase order.

3. A Case Study on Data Modeling

- At this point, enough new information has been introduced into the problem, and we should begin structuring it. There is a relationship between VENDOR, PURCHASE_ORDER, and PO_DETAIL, with which we have become familiar. However, we are now seeing new pieces of data on the purchase order: the line item amount, the estimated freight charge, the company contact, the contact's telephone number, and the vendor's item number for the material being ordered. What do we do with this new information? Let us take each item in its order.

3. A Case Study on Data Modeling
3.1 Business Process Description

The purchasing process works as follows:

- When materials are needed, an employee initiates a purchase order.
- Each purchase order has a unique PO number, a release date, status (pending, approved, completed), total amount, and estimated freight costs.
- The employee who creates the purchase order serves as the contact person for that order.
- Each purchase order is sent to exactly one vendor, but a vendor may receive many purchase orders over time.
- A purchase order contains one or more line items (purchase order details).
- Each line item specifies a particular material, quantity ordered, unit cost, promised delivery date, current balance (quantity still due), and status.
- The company maintains a list of approved materials. Each material has a unique ID, description, and unit of measure (e.g., kg, each, box).
- The company also maintains a list of approved vendors. Each vendor has a unique ID, name, address information (street, city, state, zip), phone number, fax number, and a primary contact person.
- Each vendor can supply multiple materials, and each material can be supplied by multiple vendors.
- The company tracks which vendors supply which materials, along with the vendor's item code for each material they supply.
- The company maintains employee information including ID, name (last, first, middle), home phone, department, department phone, and address information.